to incorporate transaction costs into the investment decision. Tutuncu and Koenig (2003) address the optimal asset allocation problem under the scenario where the estimated returns are unreliable. Balduzzi and Lynch (1999) study a multi-period optimization problem in the presence of costs that are either fixed or proportional to trade value. Malamut and Kissell (2002) study efficient implementation of a multi-period trade cost optimization from the perspective of the trader. Mitchell and Braun (2004) also consider portfolio rebalancing in the presence of convex transaction costs where costs are dependent solely on the quantity of shares transacted. Engle and Ferstenberg (2007) discuss a cost-adjusted frontier.

Most of these research works fall into what we coined the second wave of portfolio optimizers earlier in the chapter. In this section, we introduce the necessary techniques to solve the portfolio optimization problem (third wave of optimizers) and determine the best execution frontier. We differentiate from the above works in many ways (see Kissell & Malamut, 2007). For example:

1. We examine the portfolio optimization in terms of both market impact cost and trading risk.
2. We define market impact to be dependent upon the size of the order and the underlying execution strategy. In this case, investors have the opportunity to achieve further cost reduction through trading a diversified and/or well-hedged portfolio. Thus, depending upon the underlying trade list, the number of shares in an order could vary.
3. Our solution is based on a multi-period optimization problem that separates the total investment horizon into a trading period where shares are transacted and a holding period starting after the acquisition of the targeted portfolio and where no other shares are transacted. These problems are linked by the total shares to trade S and the trade schedule used to acquire those shares, that is, $S = \sum x$.
4. Our ultimate goal in this section is the quest for the best execution frontier.

An interesting aspect of the current portfolio construction environment is that the process appears to be backwards. For example, the results of the current portfolio optimizer provide us with a targeted future portfolio. The next question is to determine how we should best get to that end point. But what if the road is too bumpy or if no efficient road exists? Then what? We would only find this out after setting out on our journey.

A better process is a forward looking view of portfolio construction. Rather than start with the future targeted portfolio and work our way backwards through the unknown as we do now, we begin to move